Akshay Jaiswal

San Jose, CA | +1 (562) 615-1306 | jaiswalakshay57@gmail.com | linkedin.com/in/akshayjaiswal01 | github.com/Jaiswal-Akshay

AI/ML & Software Engineer | 4+ Years in Cloud Systems | Generative AI, RAG, Computer Vision, AI Agents, AWS & Docker

Education

Master of Science in Computer Science | California State University, Long Beach | GPA: 3.667 **Jan 2025 – Dec 2026**

Bachelor of Engineering in Computer Science | Rajiv Gandhi Proudyogiki Vishwavidyalaya, India **Aug 2016 – Jun 2020**

Professional Experience

Software Engineer | Tata Consultancy Services| Mumbai, India **Nov 2020 – Jan 2025**

- Led the migration of BMO's legacy on-premises banking systems to AWS, improving scalability, achieving 99.9% uptime, and reducing infrastructure costs by 25% while maintaining security and regulatory compliance.
- Engineered and optimized cloud infrastructure and CI/CD pipelines, reducing deployment time by 40% and increasing release frequency from weekly to daily builds.
- Drove incident response and resolution, handling 50+ production incidents/month and reducing MTTR by 30% through root cause analysis and cross-team coordination.

Intern | Tata Consultancy Services| Nagpur, India **Jan 2020 – Jul 2020**

- Designed and developed RESTful APIs for an employee management system, and improving data retrieval efficiency by $\sim$ 35%.
- Built backend services using Java, Spring Framework, Maven, and Tomcat to enable CRUD operations and streamline workflows, reducing manual data handling effort by 40%.

Projects

FinNext | RL-Based Personalized Financial Recommendation System **Oct 2026 – Present**

- Developed a personalized financial recommendation engine using PyTorch, Transformers, and contextual bandits, targeting a 20% improvement in recommendation acceptance over rule-based baselines.
- Implemented Conservative Q-Learning (CQL) to optimize sequential financial decisions, targeting a 15% increase in cumulative reward and a 12% reduction in simulated cash shortfalls.
- Built a low-latency recommendation API using FastAPI and Docker, targeting sub-100 ms inference latency, with MLflow for experiment tracking and reproducible model evaluation.

Research Paper Assistant & Codebase Intelligence Assistant | Local RAG Systems **Sept 2026 – Oct 2026**

- Built local RAG assistants for academic research and codebase analysis, supporting up to 10,000+ arXiv papers per topic.
- Implemented semantic retrieval with LangChain, Sentence Transformers, and ChromaDB to retrieve the top 8 evidence sources per query and generate cited responses.
- Integrated Ollama and Llama 3 for 100% local inference and developed an interactive Streamlit interface, eliminating paid LLM API dependency.

ChestX-AI Assistant(Research Prototype) &Pneumonia Detection **Jan 2026 – Jun 2026**

- Developed Computer Vision and Transfer Learning models (DenseNet-121, ResNet-18) for multi-label classification of 5 thoracic diseases using 191K+ chest X-ray images from 45K+ patients, achieving 0.75+ mean AUROC.
- Designed a multi-model, multi-agent AI pipeline integrating Groq-hosted LLMs with computer vision models to transform diagnostic probabilities into structured radiology reports, reducing manual interpretation effort and improving report consistency.
- Built an evidence-grounded Agentic AI architecture leveraging Prompt Engineering, Explainable AI (Grad-CAM), and structured JSON reasoning to generate 4-section clinical reports while mitigating hallucinations and supporting clinical decision-making.

QuizLet | CSU AI Hackathon (IBM SkillsBuild) **Mar 2026**

- Developed an AI-powered learning assistant within 24 hours during the CSU AI Hackathon, leveraging Generative AI, Prompt Engineering, IBM Watson, IBM Orchestrate, and Ollama to automate quiz generation and personalized learning workflows.
- Collaborated with a cross-functional team to build and demo a functional prototype for the Education & Workforce Intelligence track, applying Machine Learning, Intelligent Automation, Streamlit, Python, and Responsible AI principles under IBM mentorship.

Technical Skills

Languages: Python, Java, JavaScript, C/C++, C#, PHP

Machine Learning & Computer Vision: PyTorch, Transformers, Reinforcement Learning, (CQL), Contextual Bandits, Recommendation Systems, CNNs, GANs, ResNet, OpenCV, NumPy, Pandas, Matplotlib, CUDA, PyTorch Profiler, NVIDIA Nsight

Generative AI & LLMs: RAG, LangChain, LangGraph, Multi-Agent Systems, MCP (Model Context Protocol), Hugging Face, Sentence Transformers, Ollama, Llama 3, Semantic Search, Prompt Engineering

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, RDS, API Gateway, CloudFormation, CloudWatch), Docker, FastAPI, MLflow, Terraform, Ansible, CI/CD (GitHub Actions, Jenkins, AWS CodePipeline)

Databases: PostgreSQL, Oracle, SQL, DynamoDB, ChromaDB, Pinecone

Tools: Git, GitHub, Bitbucket, Jira, ServiceNow, Jupyter